Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Phase 2, Multicenter, Open-Label Study of CC-97540 (BMS-986353), CD19-Targeted NEX-T Chimeric Antigen Receptor (CAR) T Cells, in Participants With Active SLE (Including Lupus Nephritis) With Inadequate Response to Glucocorticoids and at Least 2 Immunosuppressants **Short Title/ Acronym:** A Study of CC-97540 (BMS-986353), CD19-Targeted NEX-T CAR T Cells, in Participants With Active SLE Despite Immunosuppressants (Breakfree-SLE) **Protocol Number:** CA061-1011 **NCT Number:** NCT07015983 **Posting ID:** 60938916653917292 **Sponsor/ Company Name:** Bristol Myers Squibb **Investigational Product:** CC-97540 (BMS-986353 or Zola-cel), CD19-Targeted NEX-T CAR T Cells (Preferred UMLS Name: CD19-Directed Chimeric Antigen Receptor T-Cell Therapy) **Phase of Development:** Phase 2 **Medical Monitor:** BMS Clinical Trials Contact Center / Investigator

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2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy, safety, and drug levels of CC-97540 in participants with active systemic lupus erythematosus (SLE), including lupus nephritis, who have an inadequate response to glucocorticoids and at least 2 immunosuppressants. * **Secondary Objectives:** To evaluate the drug-free achievement of Definition of Remission in SLE (DORIS); complete renal response (CRR) for participants with baseline lupus nephritis (LN); reduction in SLEDAI-2K and Physician Global Assessment (PGA) scores; and to monitor B-cell depletion and CAR T cell expansion.

2.2 Background & Rationale To address the significant medical necessity for severe, refractory SLE patients who fail standard therapies. Despite the use of glucocorticoids and chronic immunosuppressants, many patients retain active disease. CC-97540 (a CD19-directed CAR T cell therapy) aims to drive an "immune reset" and achieve complete peripheral B-cell depletion, leading to sustained, drug-free disease control and remission.

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3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm * **Blinding:** Open-label * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Target N:** 89 participants * **Number of Sites:** Multicenter (Global sites, including the US and

UK) * **Trial Status:** Recruiting * **Estimated Primary Completion:** Follow-up duration extends up to 60 months.

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants must meet the EULAR/ACR 2019 criteria for SLE. 2. Participants must have an inadequate response to appropriate doses of glucocorticoids and ≥ 2 immunosuppressant therapies, used for at least 3 months. 3. Participants must have active disease (including lupus nephritis) when signing the Informed Consent Form (ICF).

4.2 Exclusion Criteria 1. Participants must not have other diseases, conditions, or treatments that may confound interpretation of the effects of CC-97540 in SLE. 2. Uncontrolled or clinically significant cardiovascular conditions, CNS pathology, or a prior history of malignancies/lymphoproliferative disease (unless disease-free for ≥ 2 years, except for some non-invasive malignancies). 3. Prior treatment with CAR T cell therapy, genetically modified T cell therapy, stem cell transplant, or organ transplant. Washout periods may be required. 4. Received live vaccines within 6 weeks before CC-97540 administration. 5. Individuals of childbearing potential (IOCBP) who are pregnant, nursing, breastfeeding, or intend to become pregnant during the study. 6. Participant must not have inadequate organ function.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Single infusion of BMS-986353 at 10×10^6 CAR+ T cells after lymphodepletion. * **Route of Administration:** Intravenous (IV) Infusion. * **Duration of Treatment:** Single administration with a long-term follow-up of 60 months.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Lymphodepleting chemotherapy (LDC) consisting of fludarabine and cyclophosphamide (ATC Code: L01AA01; Trade names: Cytoxan, Endoxana) prior to CAR T cell infusion. * **Prohibited:** Live vaccines within 6 weeks before LDC; concurrent therapies that might confound interpretation of CC-97540 efficacy.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to 0 | ICF signing, Eligibility verification, SLEDAI-2K baseline scoring | | **Baseline/LDC**| Day -5 to Day -2 | Lymphodepleting Chemotherapy (Fludarabine/Cyclophosphamide) | | **Infusion** | Day 0 | Single IV Infusion of CC-97540 (BMS-986353) | | **Interim** | Month 6 |

Efficacy Assessment (DORIS achievement), SLEDAI-2K/PGA scoring, Safety Labs | | **End of Study**| Month 60 | Long-term follow-up (up to 5 years), final safety and remission status |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Proportion of participants achieving drug-free Definition of Remission in Systemic Lupus Erythematosus (DORIS) remission. * **Measurement Method:** Clinical assessment and laboratory analysis.

7.2 Secondary & Exploratory Endpoints * Complete renal response (CRR) for participants with baseline lupus nephritis (LN). * Participants with drug-free DORIS remission. * Safety and tolerability profile (incidence of Cytokine Release Syndrome [CRS], ICANS, cytopenias, etc.). * Changes from baseline in SLEDAI-2K and Physician Global Assessment (PGA) scores. * Pharmacodynamics/Pharmacokinetics: CAR T cell expansion, complete peripheral B-cell depletion, and reduction in Anti-dsDNA autoantibody titers.

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8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Intensive monitoring post-infusion for transient/reversible cytopenias (e.g., neutropenia, anemia), infections, and nausea.
- **Serious Adverse Events (SAEs):** Specific protocols for reporting Cytokine Release Syndrome (CRS) and Immune effector cell-associated neurotoxicity syndrome (ICANS).
- **Data Monitoring Committee (DMC):** Evaluates ongoing safety, dose-limiting toxicities (DLTs), and guided the selection of the Recommended Phase 2 Dose (RP2D) based on optimal interval design.

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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intention-to-Treat (ITT) for primary efficacy (DORIS) and Safety Set for treatment-emergent adverse events (TEAEs).
- **Sample Size Justification:** 89 participants to sufficiently evaluate the primary efficacy endpoint and map the safety profile of the RP2D.
- **Handling of Missing Data:** Standard handling methodologies per the sponsor's SAP, typically utilizing LOCF or mixed-model repeated measures for continuous clinical variables.

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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
- **Monitoring Plan:** Regular onsite and remote monitoring to ensure protocol adherence, subject safety, and data integrity.
- **Audit Trail:** eCRF locking, tracking of grading for CRS/ICANS (ASTCT consensus guidelines), and standard quality audit mechanisms.

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11. Study Identifiers & Governance

Primary ID: CA061-1011 **Posting ID:** 60938916653917292 **Registry Number:** NCT07015983 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** Breakfree-SLE **Source Level:** 5

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12. Clinical Framework (SLE Specific)

Disease Areas: Lupus Erythematosus, Systemic, Lupus Nephritis **Common Name:** Lupus **Preferred Name:** Lupus Erythematosus, Systemic **Semantic Type:** Disease or Syndrome **Clinical Definition:** A chronic, relapsing, inflammatory, and often febrile multisystemic disorder of connective tissue, characterized principally by involvement of the skin, joints, kidneys, and serosal membranes. It is of unknown etiology, but is thought to represent a failure of the regulatory mechanisms of the autoimmune system. The disease is marked by a wide range of system dysfunctions, an elevated erythrocyte sedimentation rate, and the formation of LE cells in the blood or bone marrow. **Coding Identifiers:** * MeSH Code: D008180 * HPO Code: HP:0002725 * SNOMED ID: 55464009 **Medical Methodology:** CD19-Targeted NEX-T CAR T Cell therapy combined with lymphodepleting chemotherapy **Optimization Target:** Deep immune reset, B-cell repopulation, and sustained drug-free clinical remission

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13. Study Procedures & Methodology

Management Pathway: Lymphodepletion followed by specialized adoptive cell transfer **Education Protocol (HCP):** Comprehensive site training on CAR T cell handling, infusion protocols, and AE management (CRS/ICANS) **Education Protocol (Patient):** Education on infection risks, post-infusion monitoring requirements, and long-term follow-up **Quality Audit Mechanism:** Continuous monitoring of TEAEs, CAR+ T cell kinetics, and B-cell aplasia parameters

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14. Observation & Follow-Up Schedule

Total Duration: 60 months (5 years) **Onsite Visit Cadence:** Frequent acute monitoring within the first 90 days post-infusion, followed by standardized long-term visits **Remote Monitoring Frequency:** Regular long-term safety check-ins post acute phase **Checklist Review Interval:** Continuous tracking of SLEDAI-2K, PGA, and autoimmune markers (Anti-dsDNA)

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Insert Name]	____	**[DD-MMM-YYYY]**
Clinical Operations Lead	**[Insert Name]**	___	***[DD-MMM-YYYY]***
Lead Statistician	***[Insert Name]***	_____	[DD-MMM-YYYY]